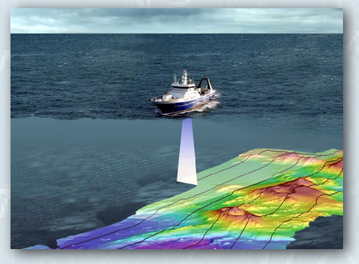
UNCHARTED WATERS

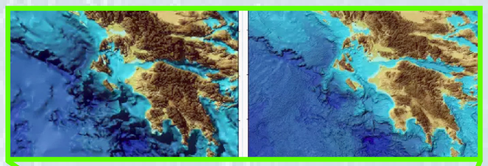
An exploration of progress in seafloor mapping

EXPLORING THE DEPTHS
Bathymetry (water depth) data has played a critical role in safe navigation since the earliest days of sailing. On the high seas, accurate depth measurements can mean the difference between safe passage or an unexpected and tragic sinking. To this day, the study of water depths and the shape of the seafloor remains one of the most important parts of voyaging.

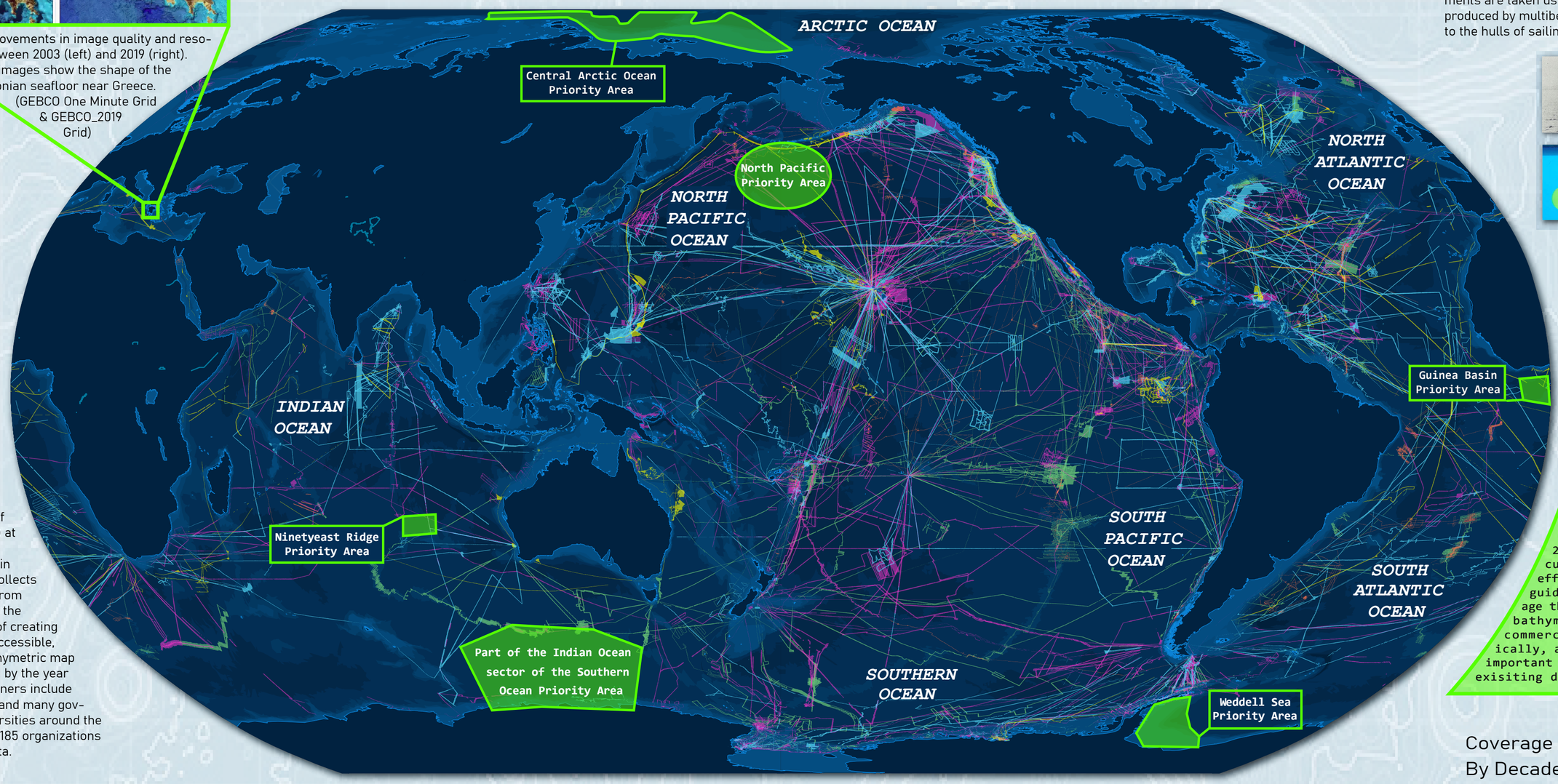
Though satellites can produce general maps of our ocean floors, the details and contours of the murky depths are obscured by waves and lack of light. Since the year 2000 C.E., at least two U.S. submarines have collided with underwater mountains called seamounts, caused in part because the seamounts had previously been uncharted. So how do we best explore the ocean floor and avoid future catastrophic incidents? The key to unlocking the deep lies in capturing detailed bathymetry data with ship-based multibeam sonar.



COLLECTING BATHYMETRY DATA
In the past, depth measurements were taken by dropping a weighted line over the side of the ship and measuring the length of line it took to reach the bottom. Today, measurements are taken using sound waves produced by multibeam sonars fixed to the hulls of sailing vessels.



Note the improvements in image quality and resolution between 2003 (left) and 2019 (right). Both images show the shape of the Ionian seafloor near Greece. (GEBCO One Minute Grid & GEBCO_2019 Grid)



MILESTONE
As of June 2025, 27.3% of the ocean floor has been mapped to modern standards! This is thanks in large part to the efforts of Seabed 2030, a global initiative launched by the Nippon Foundation and the General Bathymetric Chart of the Oceans (GEBCO) at the United Nations Oceans Conference in 2017. The initiative collects data and expertise from contributors around the world with the aim of creating a complete, freely accessible, high-resolution bathymetric map of the global seabed by the year 2030 C.E. Major partners include NOAA, ESRI, Fugro, and many governments and universities around the world. To date, over 185 organizations have contributed data.

PRIORITY AREAS
Priority areas are regions outlined by Seabed 2030 to guide current mapping efforts. These guidelines encourage the collection of bathymetry data in commercially, ecologically, and politically important areas where exisiting data is sparse.

Coverage Paths By Decade

- 2020 - 2025
- 2010 - 2019
- 2000 - 2009
- 1990 - 1999
- Before 1989

2,000 Kilometers

MODERN BATHYMETRY MAPPING
On the left is the Heezen-Tharp "World Ocean Floor" map, painted in 1977. This map is the result of a lifetime of research done by 20th century cartographer Marie Tharp and her partner Bruce Heezen. Among other extraordinary achievements, Tharp used single-beam sonar data collected by the U.S. Navy to chart the location of the Mid-Atlantic Ridge, thus proving the idea of continental drift and illustrating that the ocean floor is not all smooth and flat (as was the general belief at that time). Tharp's discoveries set the standard for maps displaying topograpy, bathymetry, and shoreline data such as the 2022 ETOPO Global Relief Model, shown above on the right.

PAST COVERAGE PATHS AND FUTURE PRIORITY AREAS
The above map shows **multibeam bathymetry coverage paths** (the voyage routes of ships that took multibeam bathymetry scans) dating back to 1970. Marine vessels equipped with multibeam sonars scan the ocean floor, producing detailed measurements of both depth and seabed shape (similar to topography on land). Many of the paths overlap, allowing researchers to verify and update existing data. Significant progress has been made, particularly since the early 2000s. **More of the ocean floor has been mapped to modern standards than ever before**; however, note the marine areas on the map without any coverage paths. These are the areas that have yet to be mapped in detail - over 60% of the ocean floor in total! Current efforts are focused on **priority areas** identified by the Nippon Foundation-GEBCO Seabed 2030 Project. These priority areas are selected using strategic rationale and include areas such as the Weddell Sea and regions in the North Pacific and Indian Oceans (several approximate areas have been highlighted in green above).

Cartographer: Hope McBride | 12/2025 | Projected Coordinate System: Robinson (world) | Central Meridian: 180 | Central Map Scale: 174,000,000 | Data Sources: Esri, TomTom, Garmin, FAO, NOAA, USFS, (c) OpenStreetMap contributors, and the GIS User Community; Bathymetric Coverage Paths: Ryan, W.B.F., S.M. Carbotte, J.O. Coplan, S. O'Hara, A. Melkonian, R. Arko, R.A. Weissel, V. Ferrini, A. Goodwillie, F. Nitsche, J. Bonczkowski, and R. Zemsky (2009), Global Multi-Resolution Topography synthesis, Geochem. Geophys. Geosyst., 10, Q03014, doi: 10.1029/2008GC002332; Priority Areas, Seabed 2030: https://seabed2030.org/our-priority-areas/; Seabed 2030 Logo: https://seabed2030.buzzsprout.com/; Ship Image, US Naval Institute: https://www.usni.org/magazines/proceedings/2020/march/mapping-world-ocean-seafloor; NOAA 2022 Global Relief Map: https://www.ncei.noaa.gov/sites/default/files/2021-10/color_etopo1_ice_low.jpg; NOAA Multibeam Sonar Image: https://nauticalcharts.noaa.gov/learn/hydrographic-survey-equipment.html; Library of Congress Blog, Marie Tharp: https://blogs.loc.gov/loc/2021/08/marie-tharp-mapping-the-ocean-floor/; Leadline Sounding Image: https://www.noaa.gov/digital-collections/collections/photo-library/4762/item?page=1; Ionian Sea Comparison, GEBCO: https://www.gebco.net/data-products/historical-data-sets; Background: GarryKillian, Vector abstract moon relief map generated conceptual lunar elevation map isolines of landscape surface elevation geographic map conceptual design elegant background for presentations, freepik.com; North Icon adapted from"bathymetry" by Tamrin at the Noun Project